

NYU, Tandon School of Engineering

Bridge to CS

Homework #3

Submission Instructions

1. For this assignment, you should turn in **3 .cpp files**, one for each coding question (Questions 1–3).
2. Name your files:
 - hw3_q1.cpp
 - hw3_q2.cpp
 - hw3_q3.cpp
3. You should submit your homework in the Gradescope system.
4. For the coding questions, pay special attention to the style of your code: indent correctly, choose meaningful variable names, define constants where needed, and choose the most appropriate control flow statements.
5. **Tip:** For this homework, it is often a good idea to **avoid writing additional helper functions** and to **avoid using loops (e.g., for-loops)** unless the problem explicitly calls for them.

Question 1

Write a program that asks the user for the length and width of a rectangle (as integers) and prints:

- The perimeter ($2 \times (\text{length} + \text{width})$)
- The area ($\text{length} \times \text{width}$)

Your program should interact with the user **exactly** as it shows in the following example:

```
Enter the rectangle length: 7
Enter the rectangle width: 4
Perimeter: 22
Area: 28
```

Question 2

Write a program that asks the user to enter an integer and prints whether it is **Even** or **Odd**.

Hint: Use the % operator.

Your program should interact with the user **exactly** as it shows in the following example:

```
Please enter an integer: 19
19 is Odd
```

Question 3

Write a program that asks the user for three integers and prints the **largest** of the three numbers.

Note: Do not use loops.

Your program should interact with the user **exactly** as it shows in the following example:

```
Please enter three integers separated by spaces:  
12 5 12  
The largest value is: 12
```